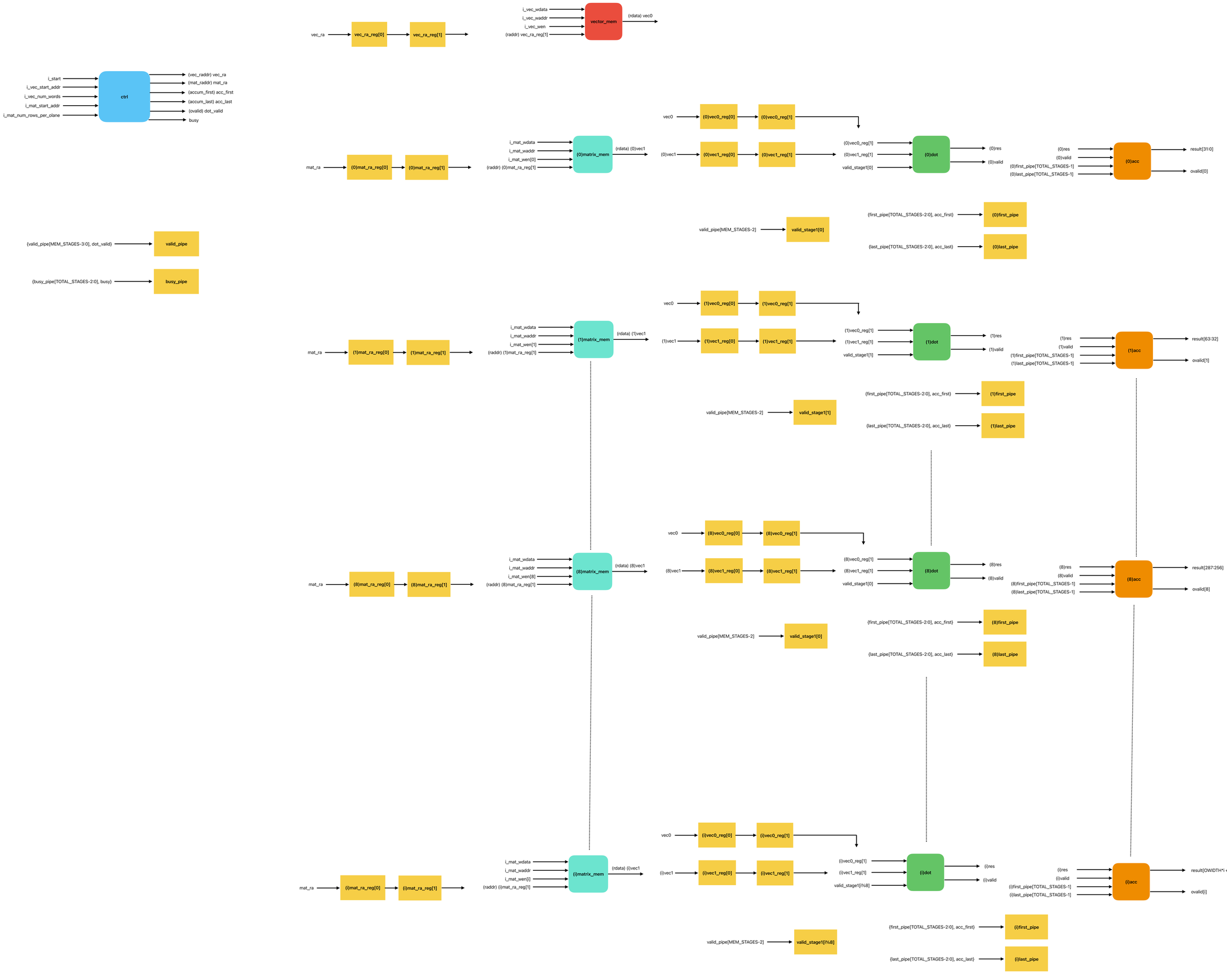
vec_ra
vec_ra_reg[0]
vec_ra_reg[1]
i_vec_wdata
i_vec_waddr
i_vec_wen
(raddr) vec_ra_reg[1]
vector_mem
(rdata) vec0
i_start
i_vec_start_addr
i_vec_num_words
i_mat_start_addr
i_mat_num_rows_per_olane
ctrl
(vec_raddr) vec_ra
(mat_raddr) mat_ra
(accum_first) acc_first
(accum_last) acc_last
(ovalid) dot_valid
busy
mat_ra
(0)mat_ra_reg[0]
(0)mat_ra_reg[1]
i_mat_wdata
i_mat_waddr
i_mat_wen[0]
(raddr) (0)mat_ra_reg[1]
(0)matrix_mem
(rdata) (0)vec1
vec0
(0)vec0_reg[0]
(0)vec0_reg[1]
(0)vec1
(0)vec1_reg[0]
(0)vec1_reg[1]
(0)vec0_reg[1]
(0)vec1_reg[1]
valid_stage1[0]
(0)dot
(0)res
(0)valid
(0)res
(0)valid
(0)first_pipe[TOTAL_STAGES-1]
(0)last_pipe[TOTAL_STAGES-1]
(0)acc
result[31:0]
ovalid[0]
{valid_pipe[MEM_STAGES-3:0], dot_valid}
valid_pipe
{busy_pipe[TOTAL_STAGES-2:0], busy}
busy_pipe
valid_pipe[MEM_STAGES-2]
valid_stage1[0]
{first_pipe[TOTAL_STAGES-2:0], acc_first}
(0)first_pipe
{last_pipe[TOTAL_STAGES-2:0], acc_last}
(0)last_pipe
mat_ra
(1)mat_ra_reg[0]
(1)mat_ra_reg[1]
i_mat_wdata
i_mat_waddr
i_mat_wen[1]
(raddr) (1)mat_ra_reg[1]
(1)matrix_mem
(rdata) (1)vec1
vec0
(1)vec0_reg[0]
(1)vec0_reg[1]
(1)vec1
(1)vec1_reg[0]
(1)vec1_reg[1]
(1)vec0_reg[1]
(1)vec1_reg[1]
valid_stage1[1]
(1)dot
(1)res
(1)valid
(1)res
(1)valid
(1)first_pipe[TOTAL_STAGES-1]
(1)last_pipe[TOTAL_STAGES-1]
(1)acc
result[63:32]
ovalid[1]
valid_pipe[MEM_STAGES-2]
valid_stage1[1]
{first_pipe[TOTAL_STAGES-2:0], acc_first}
(1)first_pipe
{last_pipe[TOTAL_STAGES-2:0], acc_last}
(1)last_pipe
mat_ra
(8)mat_ra_reg[0]
(8)mat_ra_reg[1]
i_mat_wdata
i_mat_waddr
i_mat_wen[8]
(raddr) (8)mat_ra_reg[1]
(8)matrix_mem
(rdata) (8)vec1
vec0
(8)vec0_reg[0]
(8)vec0_reg[1]
(8)vec1
(8)vec1_reg[0]
(8)vec1_reg[1]
(8)vec0_reg[1]
(8)vec1_reg[1]
valid_stage1[0]
(8)dot
(8)res
(8)valid
(8)res
(8)valid
(8)first_pipe[TOTAL_STAGES-1]
(8)last_pipe[TOTAL_STAGES-1]
(8)acc
result[287:256]
ovalid[8]
valid_pipe[MEM_STAGES-2]
valid_stage1[0]
{first_pipe[TOTAL_STAGES-2:0], acc_first}
(8)first_pipe
{last_pipe[TOTAL_STAGES-2:0], acc_last}
(8)last_pipe
busy_pipe[TOTAL_STAGES-1]
o_busy
result
o_result
ovalid[0]
o_valid
mat_ra
(i)mat_ra_reg[0]
(i)mat_ra_reg[1]
i_mat_wdata
i_mat_waddr
i_mat_wen[i]
(raddr) (i)mat_ra_reg[1]
(i)matrix_mem
(rdata) (i)vec1
vec0
(i)vec0_reg[0]
(i)vec0_reg[1]
(i)vec1
(i)vec1_reg[0]
(i)vec1_reg[1]
(i)vec0_reg[1]
(i)vec1_reg[1]
valid_stage1[i%8]
(i)dot
(i)res
(i)valid
(i)res
(i)valid
(i)first_pipe[TOTAL_STAGES-1]
(i)last_pipe[TOTAL_STAGES-1]
(i)acc
result[OWIDTH*i +: OWIDTH]
ovalid[i]
valid_pipe[MEM_STAGES-2]
valid_stage1[i%8]
{first_pipe[TOTAL_STAGES-2:0], acc_first}
(i)first_pipe
{last_pipe[TOTAL_STAGES-2:0], acc_last}
(i)last_pipe